AI-SA-IN-TH-000013

Module 8.1 · LLM Evaluation Fundamentals · AI Safety: Evaluation, Guardrails & Responsible Deployment

Transparency and Explainability

LIME, SHAP, Model Cards, EU AI Act Requirements, and the Accuracy-Explainability Tradeoff

9 min Duration	**B3 – Apply** Bloom's	**L3 – Advanced** Level	**Theory** Domain

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Learning Objectives

1. Distinguish transparency from explainability and identify which is relevant for a given stakeholder
2. Explain why LLMs are particularly opaque compared to earlier ML architectures
3. Describe the LIME algorithm mechanism and interpret a LIME explanation for a text classifier
4. Explain how SHAP assigns feature importance using Shapley values from cooperative game theory
5. Design a model card for a deployed LLM covering all mandatory fields
6. Map EU AI Act Articles 13 and 52 to concrete transparency obligations for LLM developers

Transparency vs Explainability — The Distinction

Transparency vs Explainability

Transparency refers to openness about how an AI system is designed, trained, and deployed — the 'what and how' of the system's construction. Explainability refers to the ability to describe, in human-understandable terms, why a specific AI decision or output was produced — the 'why' of individual decisions.

- Transparency: publishing training data details, model architecture, evaluation results, known limitations — answers 'what is this system'
- Explainability: providing a reason for why loan application X was rejected, why image Y was classified as a cat — answers 'why did it decide this'
- Interpretability: a related term — a model is interpretable if a human can understand its internal mechanics directly (e.g. decision tree); LLMs are not interpretable but can be made partially explainable via XAI tools
- Stakeholder split: regulators need transparency (auditable systems); users need explainability (decisions about them); developers need interpretability (debugging)
- Legal obligation: GDPR Article 22 mandates explainability for automated decisions; EU AI Act Article 13 mandates transparency documentation for high-risk AI

Black-Box vs White-Box AI Models

White-Box (Interpretable) Models

- Internal mechanics directly understandable by humans
- Linear regression: weights show feature contribution
- Decision tree: rules can be read and audited
- Logistic regression: log-odds intuitive
- Advantage: inherently explainable — no XAI tools needed
- Disadvantage: insufficient capacity for complex tasks (NLP, vision, code)
- Examples: credit scorecard, clinical decision rules, audit compliance classifiers

Black-Box (Opaque) Models

- Internal mechanics not directly understandable
- Deep neural networks: billions of weights in nonlinear composition
- LLMs: emergent behaviour from scale — no single decision pathway
- Gradient boosted trees: 1000s of trees in combination
- Advantage: state-of-the-art performance on complex tasks
- Disadvantage: explanations are approximations via XAI tools — not ground truth
- Regulatory risk: GDPR Art 22 requires explanation for consequential decisions

Why LLMs Are Particularly Opaque

Scale	GPT-4: ~1.8 trillion parameters. No human can trace which weights caused which token.
Emergence	Capabilities arise unpredictably from scale — even creators cannot predict what the model will learn.
Distributed Representation	A concept like 'justice' is encoded across millions of neurons — there is no single 'justice neuron'.
Attention ≠ Explanation	Attention weights show which tokens were attended to, not why — high attention ≠ causal importance.
Context Sensitivity	Same prompt can produce different outputs — the model's 'decision' cannot be reduced to a fixed function.

LIME — Local Interpretable Model-Agnostic Explanations

LIME (Ribeiro et al., 2016)

LIME explains any black-box model's prediction on a specific instance by fitting a simple, interpretable linear model (the explanation) in the neighbourhood of that instance. It is model-agnostic — works with any classifier — and local — explains one prediction at a time, not the model globally.

- Step 1: Perturb the input instance — create many small variations (e.g. mask words, hide superpixels)
- Step 2: Get predictions from the black-box model on all perturbed versions
- Step 3: Weight each perturbation by its similarity to the original instance
- Step 4: Fit a simple linear model on the weighted perturbed dataset
- Step 5: The linear model's coefficients become the explanation — positive = contributes to prediction, negative = works against it
- Limitation: explanations are local approximations — they may not faithfully represent the model's true reasoning globally

LIME — Sentiment Analysis Worked Example

Explaining why a review was classified as NEGATIVE by a sentiment classifier

Input Review (Classified as NEGATIVE, confidence 0.87)

"The hotel was clean and the staff were friendly, but the breakfast was terrible and the noise from the street made sleep impossible."

LIME Feature Importance

LIME perturbs the sentence 500 times (masking word subsets), obtains predictions, fits a local linear model.

Top contributing features:
+0.31: 'impossible' (strongly negative)
+0.28: 'terrible' (strongly negative)
−0.18: 'friendly' (positive — partially offsets)
−0.12: 'clean' (positive — partially offsets)
+0.09: 'noise' (moderately negative)

Interpretation: 'terrible' and 'impossible' are the primary drivers of the NEGATIVE classification. 'Friendly' and 'clean' are present but insufficient to overturn the negative sentiment. The explanation is locally faithful — within this neighbourhood of the input, these weights hold.

SHAP — SHapley Additive Explanations

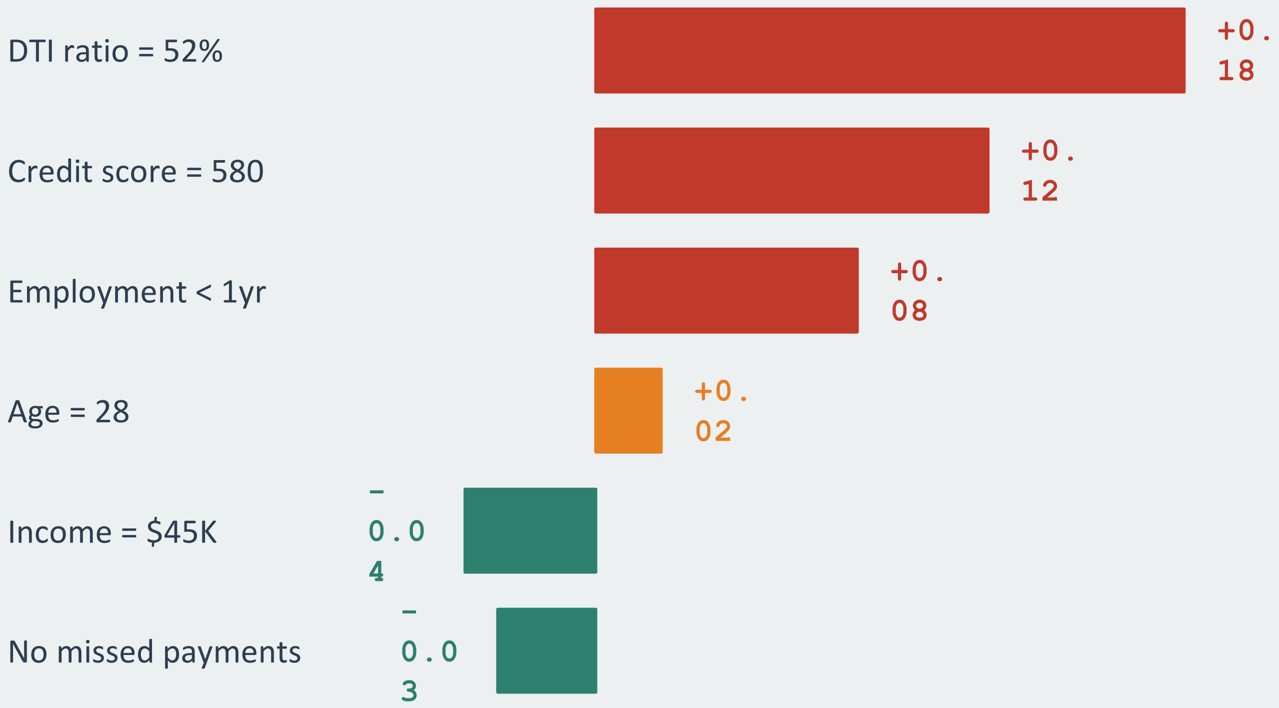
SHAP (Lundberg & Lee, 2017)

SHAP assigns each feature a contribution value (Shapley value) to a model prediction, grounded in cooperative game theory. A Shapley value represents how much each feature 'contributed' to moving the prediction away from the mean prediction — computed by averaging contributions across all possible feature coalitions.

- Game-theory grounding: borrowed from the Shapley value in cooperative games — fairly distributes credit among players
- Consistency: the only method satisfying four axioms: efficiency, symmetry, dummy, and linearity — making it uniquely fair
- Global explainability: aggregate SHAP values across a dataset to show overall feature importance
- Local explainability: explain any single prediction with the exact contribution of each feature
- SHAP for LLMs: token-level SHAP values show which input tokens most influenced a particular output token — computationally expensive but theoretically sound
- Tools: shap Python library (pip install shap); TreeExplainer for gradient boosted models; KernelExplainer for any model

SHAP Feature Importance — Visual Interpretation

Loan Application Rejected — SHAP Waterfall Plot (base value: P=0.45, final prediction: P=0.78 → REJECT)



Red = increases rejection probability · Green = decreases rejection probability · Width = magnitude

Model Cards — Design and Mandatory Fields

Model Card (Mitchell et al., 2019)

A structured documentation artifact accompanying an AI model that provides essential information about the model's intended use, performance characteristics, limitations, and potential harms — enabling stakeholders to make informed deployment decisions.

- Model details: architecture, training data, version, contact information for the developing organisation
- Intended use: primary intended uses, intended users, out-of-scope uses that should not be relied upon
- Factors: relevant demographic groups, environmental conditions, and other factors affecting performance
- Metrics: evaluation metrics used and why they were chosen, disaggregated performance across relevant subgroups
- Ethical considerations: potential risks, mitigation strategies, recommendations for deployment
- EU AI Act Article 13 requirement: transparency documentation including capabilities, limitations, and accuracy for high-risk AI systems — model cards are the standard format

Model Card — Worked Template (Medical LLM)

Template for a clinical decision support LLM — mandatory fields under EU AI Act Art 13

Model Details

Model name: MedAssist-7B · Version: 2.1 · Architecture: Llama-3.1-7B fine-tuned on PubMed + clinical notes
Organisation: Zenteiq AI · Contact: safety@zenteiq.ai · License: CC BY-NC 4.0
Intended use: Supporting physician differential diagnosis — NOT for direct patient use
Out-of-scope: Prescribing, dosing decisions, emergency triage, patient-facing deployment

Performance & Limitations

MedQA accuracy: 78.4% overall · Disaggregated by subgroup:
Age > 65: 74.1% · Paediatric: 71.3% · Rare diseases: 58.2%
Known limitations: (1) Significantly lower accuracy on rare diseases (<1:10,000); (2) Training data bias toward Western clinical guidelines; (3) Not validated on non-English clinical notes
Ethical considerations: Must always be reviewed by licensed physician before clinical action. Outputs should not be disclosed to patients directly.

Transparency Artifacts — Model Card vs System Card vs Datasheet

Artifact	What It Documents	Created By	Primary Audience	Standard
Model Card	Model capabilities, performance, limitations, intended use	Model developer	Users, regulators, downstream developers	Mitchell et al. 2019; EU AI Act Art 13
System Card	Full AI system including models, prompts, tools, APIs	System deployer	Enterprise customers, regulators, safety teams	Meta 2023 (Llama 2 system card)
Datasheet for Dataset	Dataset contents, collection methods, bias, intended uses	Dataset curator	Researchers, model trainers, auditors	Gebru et al. 2018
Algorithmic Impact Assessment	Risk assessment of AI system deployment impacts	Organisation deploying AI	Legal, compliance, ethics teams	Various national standards

EU AI Act Transparency Requirements

EU AI Act Articles 13 & 52 — Transparency Obligations

The EU AI Act (2024) imposes mandatory transparency requirements on AI systems — distinguishing between high-risk AI systems (Article 13) which require comprehensive technical documentation, and limited-risk AI systems (Article 52) which require disclosure of AI nature to users.

- Article 13 (High-Risk AI Transparency): providers must supply users with instructions for use covering intended purpose, performance, accuracy limitations, and human oversight requirements — model cards are the standard format
- Article 52 (Chatbot/Deep-fake Disclosure): AI systems interacting with natural persons must disclose they are AI; deep-fake content must be labelled — applies to all deployed LLM chatbots in the EU
- Article 9 (Risk Management): continuous risk management throughout AI system lifecycle; documentation must be updated when system changes
- Recital 47: transparency obligations scale with risk — the higher the risk, the more detailed the required documentation
- Non-compliance penalties: up to €15M or 3% of global annual turnover for Article 52 violations; up to €30M or 6% for prohibited AI practices

User-Facing vs Developer-Facing Explainability

User-Facing Explanations

- Target: end users who receive AI decisions (loan applicants, patients, job candidates)
- Form: plain language, non-technical — 'Your application was not approved because your debt-to-income ratio is above our threshold'
- Legal obligation: GDPR Article 22 — right to obtain meaningful information about the logic of automated decisions
- Design principle: should be actionable — tell the user what they could change to get a different outcome
- Pitfall: oversimplified explanations can mislead — the real model logic may be more complex than the explanation suggests

Developer-Facing Explanations

- Target: ML engineers, auditors, regulators — people who maintain or audit the system
- Form: technical — SHAP values, attention maps, decision boundaries, feature importance
- Purpose: debugging (why is the model wrong?), auditing (is the model fair?), compliance (can we justify the decision?)
- LIME and SHAP are both developer-facing tools — they require ML knowledge to interpret correctly
- EU AI Act Article 14: human oversight requires that operators 'fully understand' the AI system — developer-facing explainability is the mechanism

Case Study — GDPR Article 22 Right to Explanation in Practice

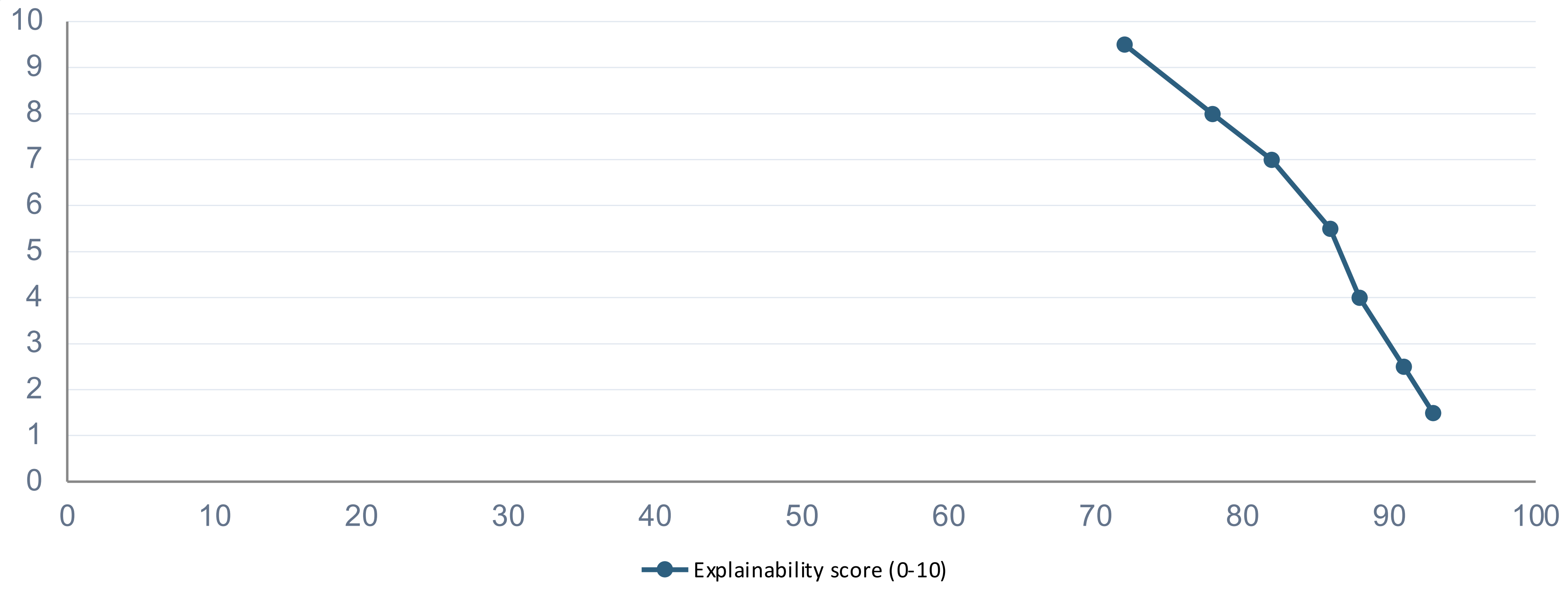
Case: UK Financial Regulator Enforcement Case (2022) — Automated Credit Decision Challenge

- A UK consumer applied for a mortgage through a major bank and was rejected by an automated underwriting AI system
- Under GDPR Article 22, the consumer requested an explanation of the automated decision — specifically, the logic underlying the rejection
- The bank provided a boilerplate response citing 'income to debt ratio criteria' — the UK ICO ruled this was insufficient under GDPR recital 71
- The ICO required the bank to provide: (1) the main factors considered, (2) their relative weights, (3) the applicant's specific values for each factor, and (4) the threshold that would have resulted in approval
- The bank's ML team had to implement post-hoc SHAP explainability for their gradient boosted model — this was not built into the system at deployment

Key Lesson

This case established that 'meaningful information' under GDPR means feature-level explanations with the applicant's specific values — not generic model descriptions. Implication: explainability must be built into AI systems before deployment, not retrofitted after a legal challenge. SHAP or LIME output must be translatable into user-facing language.

Accuracy-Explainability Tradeoff by Model Family



From left to right: Logistic Regression, Decision Tree, Random Forest, Gradient Boosted Trees, Small NN, Large Transformer, LLM-scale. As accuracy increases, native explainability decreases. XAI tools (LIME, SHAP) partially bridge the gap but provide approximations, not ground truth.

Key Takeaways

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1. Transparency (openness about system design) and explainability (reasons for individual decisions) are distinct — regulators need transparency; users need explainability; developers need interpretability

2. LLMs are opaque for five reasons: scale (trillions of parameters), emergence, distributed representations, attention ≠ causation, and context-sensitivity — XAI tools provide approximations, not ground truth

3. LIME approximates any model locally by perturbing inputs and fitting a linear surrogate; SHAP assigns Shapley values using game theory — SHAP is uniquely consistent but computationally heavier

4. Model cards are the standard transparency artifact: they document intended use, performance by subgroup, known limitations, and ethical considerations — mandatory under EU AI Act Article 13 for high-risk AI

5. EU AI Act Article 52 requires disclosure that users are interacting with AI; Article 13 requires full technical transparency documentation for high-risk systems — non-compliance fines up to €30M or 6% of global turnover